

Abhinav Reddy Kandula

Cincinnati, Ohio | kandulay@mail.uc.edu | 513-388-6340 | [linkedin.com/in/kandula-abhinav-reddy](https://www.linkedin.com/in/kandula-abhinav-reddy) | [GitHub](#)

Software Development Engineer | Backend Systems & Cloud Automation

Software Development Engineer with hands-on experience designing and building **backend services, REST APIs, and cloud-deployed applications**. Proven ability to develop **Python- and Java-based systems**, deploy **containerized applications**, and maintain **reliable, scalable server-side solutions**. Strong foundation in **data structures, algorithms, object-oriented design, operating systems, and Linux-based development**, with experience supporting production systems used by real customers.

EDUCATION

University of Cincinnati, <i>Master of Engineering in Computer Science</i>	August 2024 -May 2026
CMR Institute of Technology, <i>B. Tech in Computer Science Engineering</i>	August 2020 - June 2024

TECHNICAL SKILLS

Languages: Python, Java, C#, C++, SQL, JavaScript

Core CS: Data Structures & Algorithms, OOP, Design Patterns, Multithreading, Software Development Lifecycle (SDLC)

Backend & Systems: REST APIs, Web Services, JSON, XML, Backend Application Development

Distributed Systems (Foundations): Workflow orchestration, fault isolation, cloud-backed execution

Cloud & Data: AWS, Azure, GCP | Snowflake, Docker, Linux, CI/CD Concepts, Containerized Deployments

Databases: SQL Server, MySQL, PostgreSQL, MongoDB

ML & Computer Vision: Applied ML, Computer Vision, YOLO-based object detection, OpenCV, TensorFlow

Data & Analytics: SQL, Python (Pandas), ETL pipelines, Data validation, Reporting systems

Tools: Git/GitHub, Linux, CI/CD concepts, Docker, Kubernetes, Agile/Scrum, Automation Workflows (n8n)

EXPERIENCE

Software Development Engineer Intern – Analysis Express January 2026 – Present

- Designed and developed **Python-based backend services and APIs** for IrisX, processing large-scale engineering datasets in production environments.
- Built and deployed **containerized applications using Docker** on Linux-based systems, improving consistency across development and deployment environments.
- Designed backend workflows to support **batch processing and parallel execution**, applying multithreading concepts for performance optimization.
- Implemented **structured logging, validation, and error handling**, improving system reliability and debuggability.
- Integrated automation and orchestration workflows to manage data ingestion, processing, and export across cloud-backed systems.
- Worked with a large, multi-component codebase, contributing to iterative releases and production stability improvements.

Data Analyst & Supervisor – A&F MainStreet Operations, University of Cincinnati. September 2024 – Present

- Designed and implemented **backend automation pipelines using Python and SQL**, improving operational efficiency by 13%.
- Built and maintained **cloud-hosted reporting and analytics services** supporting real-time operational decision-making.
- Developed maintainable backend code with **robust error handling, logging, and validation** aligned with software engineering best practices.
- Led and coordinated a small technical team, owning production workflows, troubleshooting, and continuous improvements.

PROJECTS

Backend API & Data Processing System (Python)

Python, REST APIs, Docker, Linux

- Designed and implemented backend services exposing RESTful APIs for data ingestion and processing.
- Containerized services using Docker, enabling reproducible builds and deployments.
- Applied object-oriented design principles and modular architecture to improve maintainability and scalability.

Smart Inventory Management System (ETL + Backend Application)

C#, ASP.NET Core, Razor Pages, SQL Server, Entity Framework

- Designed and built a **production-style backend web application** using ASP.NET Core.
- Implemented **RESTful APIs**, authentication, and CRUD operations following clean architecture and MVC principles.
- Designed relational schemas and optimized SQL queries to improve system performance.